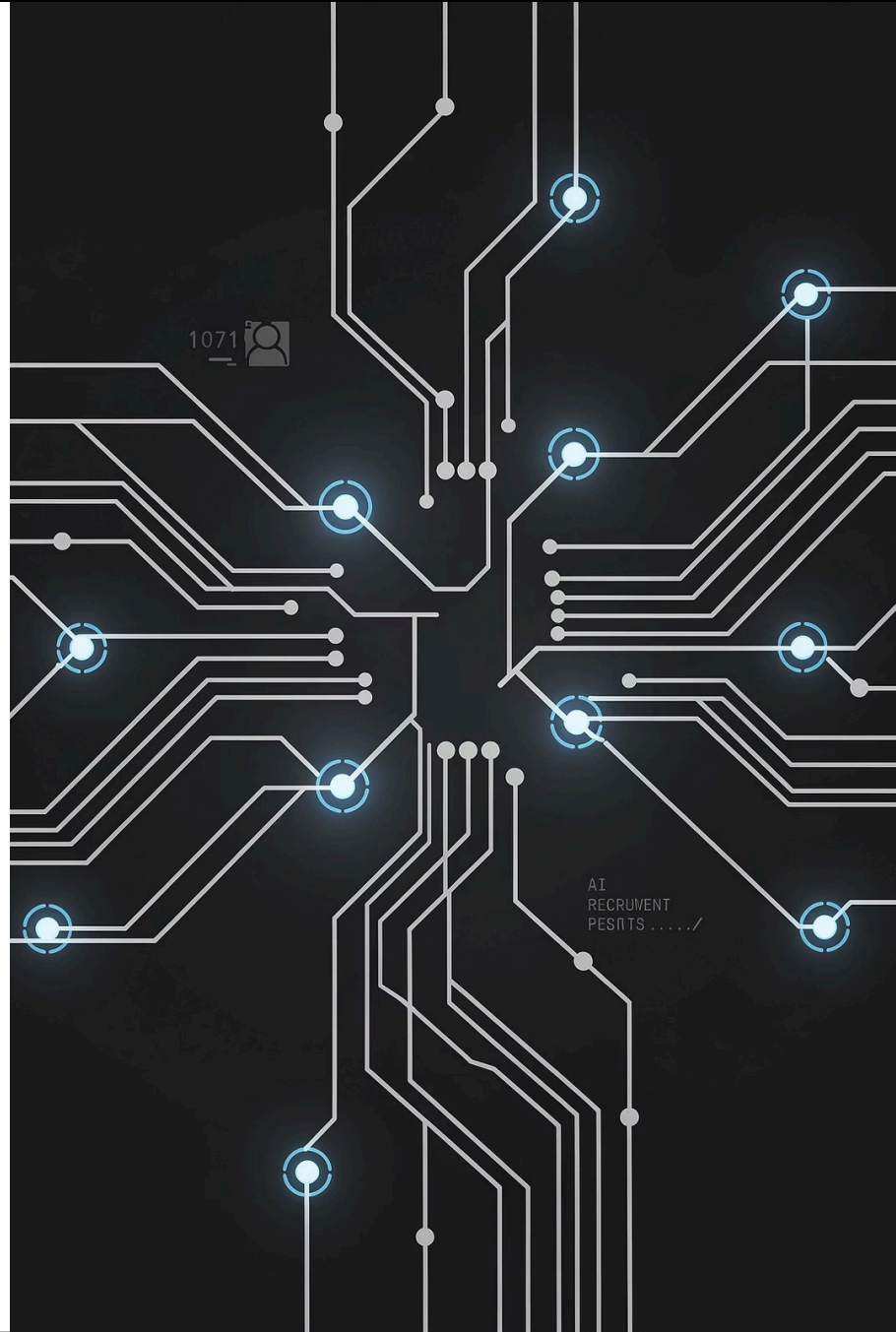
AI-Powered Candidate Matching System

Improving speed, consistency, and quality in resume screening — a smarter way to find the right talent.

Github Link

https://github.com/healthycarrot/prodapt-capstone



The Core Problem

Finding consistent, quality candidates efficiently remains recruitment's greatest challenge.

⏱️ Speed

Manual screening creates significant bottlenecks with large applicant volumes.

⚖️ Consistency

Varied evaluation criteria cause unpredictable hiring outcomes.

🎯 Quality

Keyword matching often fails to identify relevant skills accurately.

We must improve hiring speed, consistency, and quality simultaneously.

Design Thinking

We help recruiters understand candidate fit with confidence and clarity.

Semantic Understanding

Requirements are interpreted semantically, moving beyond simple keyword parsing.

Transferable Skill Recognition

Candidates match via related and transferable capabilities, not exact overlap.

Explainability First

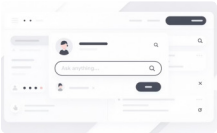
Natural language explanations support, rather than replace, human recruiter judgement.

Format-Agnostic Profiles

Diverse resume formats are normalized for consistent, fair comparison.

DECISION SUPPORT FOR RECRUITERS

Key Features

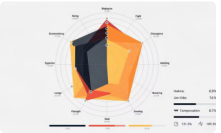


Natural Language Search

Enter hiring requirements as free text. The system semantically retrieves the most relevant candidates.

Hard Filters

Apply structured constraints — location, experience, education — to refine the candidate pool with precision.



Multi-Dimensional Scoring

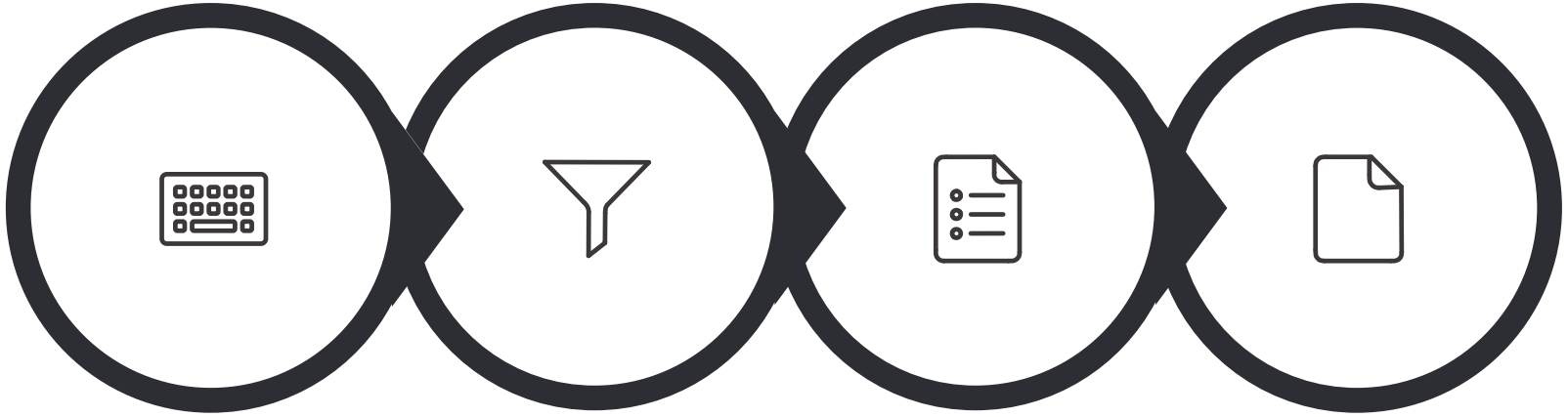
Candidates are scored across skills, experience, career progression, and soft skills — not just a single rank.

Full Candidate Access

Review natural language score explanations alongside summarised profiles and original source resumes.

Product Demo

A four-step workflow from requirement input to ranked, explained candidate results.



Enter Requirements

Apply Filters

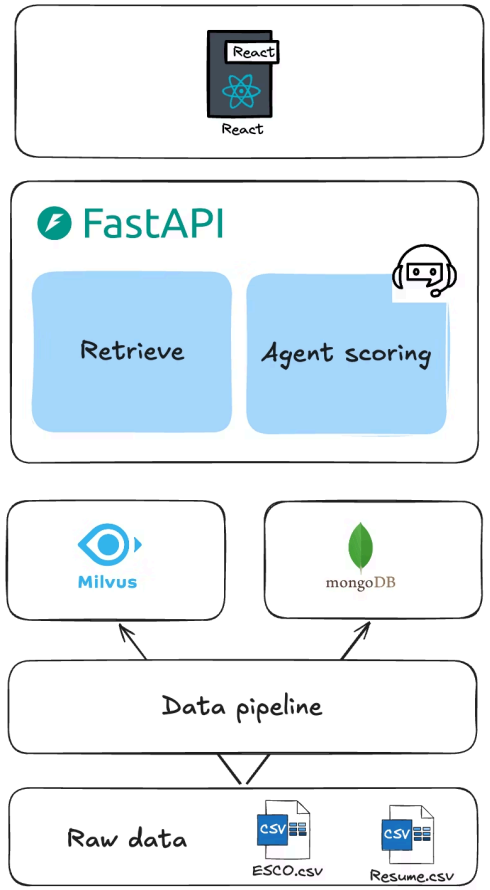
Retrieve & Rank

Review Explanations

Screenshot of the live UI will be inserted here.

ARCHITECTURE & DESIGN

Enterprise Architecture



Design Intentions

Microservice Thinking

Each capability is independently deployable and replaceable.

Separable Retrieval

The `/retrieve` capability is intentionally decoupled from the frontend search flow.

Low Switching Cost

The agent layer and vector database can be swapped without rebuilding the full system.

Modularity and future extensibility are first-class design goals.

ARCHITECTURE & DESIGN

Trade-off Decisions: Database Selection

MongoDB: System of Record

Option	Why / Decision
MongoDB	✅ Selected: Best fit for semi-structured, multi-format resumes.
PostgreSQL	❌ Rejected: Schema evolution is costly for variable fields.
Elasticsearch	❌ Rejected: Over-engineered for a system of record.

Milvus: Retrieval Engine

Option	Why / Decision
Milvus	✅ Selected: Scalable semantic search across large candidate pools.
pgvector	❌ Rejected: Not suitable for production-scale retrieval.
Pinecone / Weaviate	❌ Rejected: Concerns regarding vendor lock-in and long-term cost.

Separating MongoDB as the system of record and Milvus as the retrieval engine ensures system extensibility and high-performance semantic search.

ARCHITECTURE & DESIGN

System Thinking: Data Challenges

The Nature of This Use Case

1 Heterogeneous Sources

Resume data arrives from multiple sources in different formats — HTML, PDF, plain text, structured JSON.

2 Unstructured by Default

The majority of candidate data is unstructured, making consistent extraction non-trivial.

3 Naive Chunking Fails

Simple token-length chunking degrades retrieval quality. Passing whole documents inflates context size, increasing latency and cost.

Therefore, the **core design focus is the preprocessing pipeline** — before retrieval or scoring can be effective.

PIPELINE DESIGN

Pipeline Stage 1: Structuring

Raw resumes in varying formats are parsed and decomposed into structured, retrievable elements.

HTML Parsing

HTML resumes are parsed using semantic tags to extract sections cleanly and reliably.

NLP / LLM Parsing

Non-HTML formats use NLP-based title extraction or LLM-based parsing to identify section boundaries.

Key Field Extraction

Skill and Occupation fields are extracted as structured elements, enabling downstream RAG and filtering.

Data flow diagram for the structuring stage will be inserted here.

PIPELINE DESIGN

Pipeline Stage 2: Normalisation

The Challenge

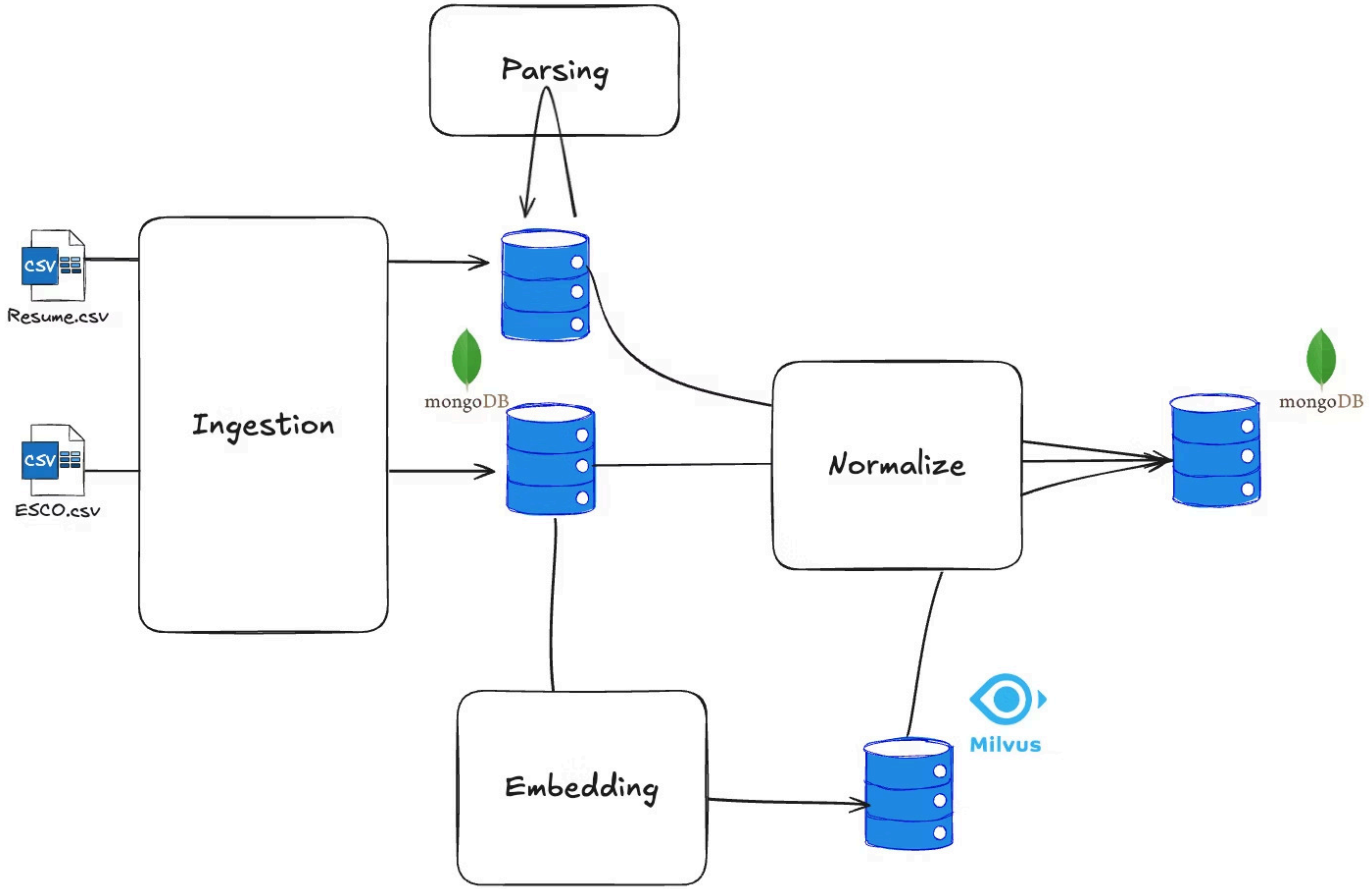
Datasets often use different terminology for identical skills and job titles.

The Approach

We normalise all candidate data to the ESCO taxonomy for consistent, standardised classification.

Why ESCO?

Use the **ESCO dataset** as the shared taxonomy for skills, competencies, qualifications, and occupations



PIPELINE DESIGN

Normalisation Methods & Impact

Three-Layer Matching

Keyword Matching

Fast exact-match for common skill terms.

Cosine Similarity

Embeds descriptions for comparison against resumes.

LLM-Based Matching

Resolves complex mappings using AI.

Impact of Normalisation

Lower Ambiguity

Reduces mismatches during retrieval.

Token Savings

Compression lowers LLM costs.

Better RAG Quality

Increases precision of results.

Extensibility

Simplified new data onboarding.

AI WORKFLOW DESIGN

Retrieval Design

A multi-stage pipeline combining filtering, hybrid search, and cross-encoder reranking.

1

Hard Filter Extraction

Filters explicit selections and normalises input to ESCO standard.

2

Hybrid Search

Runs parallel vector and keyword searches for broad, precise recall.

3

Fusion Score

Merges search paths into a single unified ranked result list.

4

Cross-Encoder Reranking

Reranks top candidates for maximum relevance before final scoring.

AI WORKFLOW DESIGN

Agent-Based Scoring

Five Specialist Agents

Skill Match

Experience Match

Education Match

Career Progression

Soft Skills

Orchestration Logic

- Specialist agents run independently and in parallel to minimize latency.
- The Orchestrator aggregates individual scores and produces a final summary.
- The Orchestrator skips unnecessary agents based on role relevance.

Business Value & ROI

⚡ Operational Efficiency

Significantly reduces time spent on manual resume screening. Lowers reviewer workload, freeing recruiters for higher-value decisions.

🎯 Hiring Quality

Surfaces strong candidates beyond keyword overlap. Identifies transferable skills that traditional screening misses.

⚖️ Consistency

Reduces evaluator variance across teams. Standardises the reasoning behind every hiring recommendation.

Measuring Success: KPIs

Operational Metrics

Average Review Time

Tracks efficiency gains by reducing time per application review.

Hiring Lead Time

Measures duration from job posting to final offer acceptance.

Rediscovery Rate

Percentage of high-quality candidates identified from previous pools.

Quality Metrics

Interview Pass-Through Rate

Measures how many AI-matched candidates advance through stages.

Top-Rank Conversion Rate

Tracks success rates for candidates ranked highly by systems.

Inter-Rater Agreement

Evaluates consistency among recruiters using AI-driven insights.

Evaluate system success through measurable impact on hiring efficiency and quality.

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